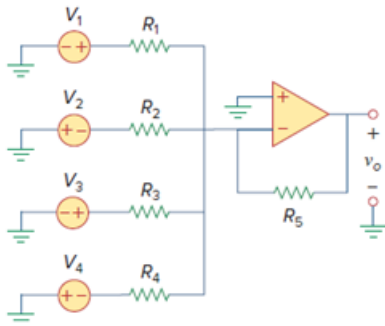


Problem

Using Fig. 5.75, design a problem to help other students better understand summing amplifiers.

Figure 5.75:



Step-by-step solution

Step 1 of 4

Refer to Figure 5.78 in the textbook.

Draw the circuit diagram with node notations.

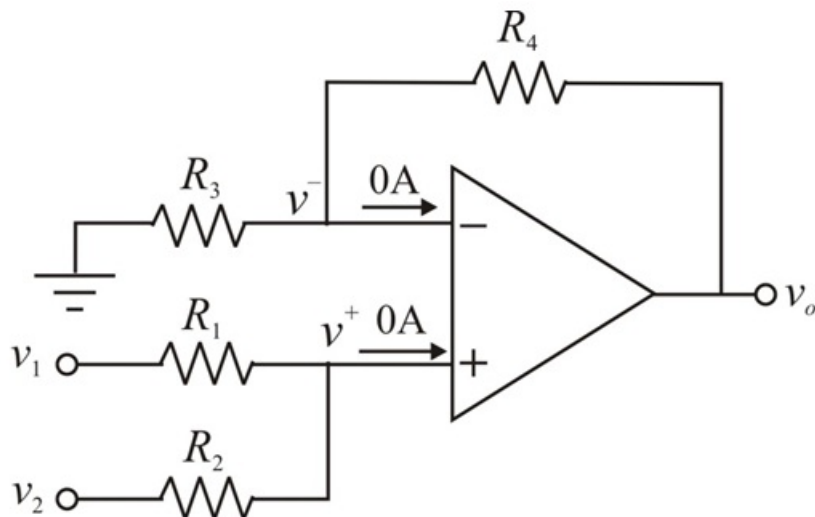


Figure 1

Step 2 of 4

In an ideal op-amp, the inverting and non-inverting terminal currents are zero and the inverting and non-inverting node voltages are equal.

$$v^- = v^+$$

Apply Kirchhoff's current law at node v^- .

$$\frac{v^-}{R_3} + \frac{v^- - v_o}{R_4} = 0$$

$$v^- \left[\frac{1}{R_3} + \frac{1}{R_4} \right] - \frac{v_o}{R_4} = 0$$

$$v^- \left[\frac{R_4 + R_3}{R_3 R_4} \right] = \frac{v_o}{R_4}$$

$$v^- \left[\frac{R_3 + R_4}{R_3} \right] = v_o$$

$$v^- = v_o \left[\frac{R_3}{R_3 + R_4} \right]$$

Step 3 of 4

Apply Kirchhoff's current law at node v^+ .

$$\frac{v^+ - v_1}{R_1} + \frac{v^+ - v_2}{R_2} = 0$$

$$\frac{R_2(v^+ - v_1) + R_1(v^+ - v_2)}{R_1 R_2} = 0$$

$$R_2(v^+ - v_1) + R_1(v^+ - v_2) = 0$$

$$R_2 v^+ - R_2 v_1 + R_1 v^+ - R_1 v_2 = 0$$

$$(R_1 + R_2)v^+ - R_2 v_1 - R_1 v_2 = 0$$

$$(R_1 + R_2)v^- - R_2 v_1 - R_1 v_2 = 0 \quad \left(\text{Since, } v^- = v^+ \right)$$

Step 4 of 4

Substitute $v_o \left[\frac{R_3}{R_3 + R_4} \right]$ for v^- .

$$v_o \left[\frac{R_3(R_1 + R_2)}{R_3 + R_4} \right] - R_2 v_1 - R_1 v_2 = 0$$

$$v_o \left(\frac{R_3(R_1 + R_2)}{R_3 + R_4} \right) = R_2 v_1 + R_1 v_2$$

$$v_o = \frac{(R_3 + R_4)}{R_3(R_1 + R_2)} (R_2 v_1 + R_1 v_2)$$

Therefore, the output voltage is

$$v_o = \frac{(R_3 + R_4)}{R_3(R_1 + R_2)} (R_2 v_1 + R_1 v_2).$$